

A Circular Microstrip Patch Antenna with Dual Band Notch Characteristics for UWB Applications

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Abstract—This report presents the design and analysis of a circular microstrip patch antenna with dual band notch characteristics for Ultra-Wideband (UWB) applications. The antenna is designed on FR4 substrate with 1.6mm thickness and operates in the frequency range of 3.1-10.6 GHz with dual band rejection at WiMAX (3.5-5.01 GHz) and RFID (5.9-7.24 GHz) bands. The design incorporates two inverted C-shaped slots for band rejection and an L-shaped slot for bandwidth enhancement. Simulated results demonstrate good return loss and VSWR characteristics across the operating bandwidth.

Index Terms—Circular microstrip patch antenna, UWB, dual band notch, WiMAX rejection, RFID rejection

I. INTRODUCTION

Ultra-Wideband (UWB) technology has gained significant attention for wireless communications due to its advantages including high data rates, low power consumption, and immunity to multipath interference. The Federal Communications Commission (FCC) allocated the unlicensed UWB spectrum from 3.1 to 10.6 GHz for commercial applications.

However, narrow band systems like WiMAX (3.5-5.5 GHz) and RFID (6.6-7.1 GHz) operate within this UWB spectrum, causing potential interference. To mitigate this issue, UWB antennas with band notch characteristics are essential. This report presents a circular microstrip patch antenna with dual band rejection capability designed for UWB applications.

II. ANTENNA DESIGN AND SPECIFICATIONS

A. Antenna Specifications

The key specifications of the designed antenna are presented in Table I.

TABLE I: Antenna Specifications

Parameter	Value
Substrate Material	FR4 Epoxy
Substrate Height (h)	1.6 mm
Dielectric Constant (ϵ_r)	4.3
Loss Tangent ($\tan \delta$)	0.025
Substrate Dimensions	$16.5 \times 18 \times 1.6 \text{ mm}^3$
Operating Frequency	3.1 - 10.6 GHz
Notch Band 1 (WiMAX)	3.5 - 5.01 GHz
Notch Band 2 (RFID)	5.9 - 7.24 GHz
Patch Radius (R)	5.4 mm
Ground Plane Length	4.6 mm
Feed Line Length	5.8 mm

B. Design Methodology

The antenna design incorporates three key features:

- **Circular Patch:** Provides wide bandwidth and omnidirectional radiation pattern
- **Dual Inverted C-Slots:** Creates two band notches for WiMAX and RFID rejection
- **L-Shaped Slot:** Enhances overall antenna bandwidth

The partial ground plane technique is employed to achieve UWB characteristics. The ground plane width of 4.6 mm was optimized for best return loss performance.

III. ANTENNA GEOMETRY

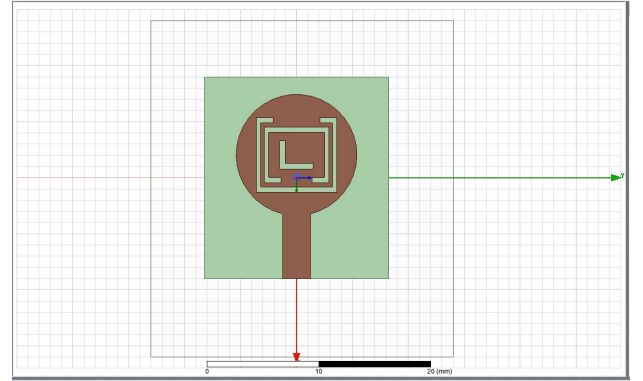


Fig. 1: Front view of the proposed antenna showing circular patch with inverted C-slots and L-slot

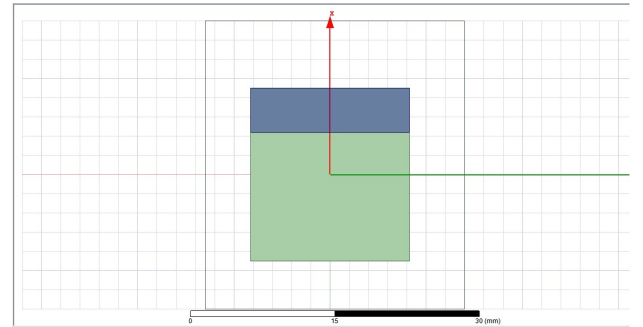


Fig. 2: Back view showing partial ground plane configuration

The circular patch is fed by a microstrip line, and the two inverted C-shaped slots are strategically positioned to achieve the desired notch frequencies.

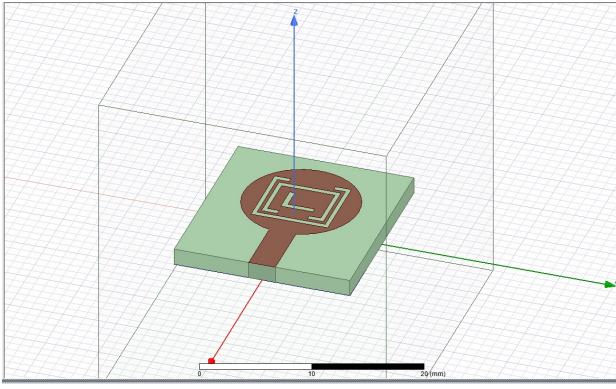


Fig. 3: Trimetric view of the antenna structure

IV. SIMULATION AND RESULTS

The antenna was simulated using HFSS (High Frequency Structure Simulator). Key performance parameters were analyzed including S11 parameter, VSWR, and radiation patterns.

A. S11 Parameter (Return Loss)

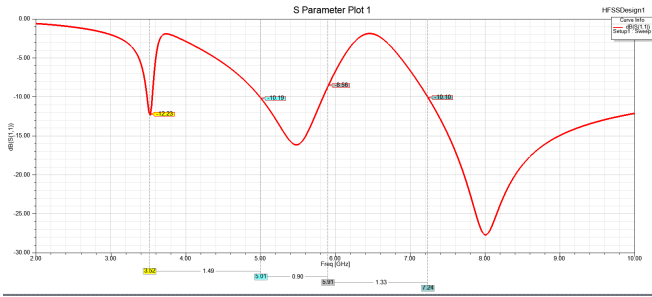


Fig. 4: Simulated S11 parameter showing dual band notches

Fig. 4 shows the simulated S11 parameter. The antenna exhibits return loss less than -10 dB across the entire UWB band (3.1-10.6 GHz) except at the two notch bands:

- Notch Band 1: 3.5-5.01 GHz (WiMAX rejection)
- Notch Band 2: 5.9-7.24 GHz (RFID rejection)

The peak return loss of -28 dB is achieved at 8 GHz, indicating excellent impedance matching at this frequency.

B. VSWR Analysis

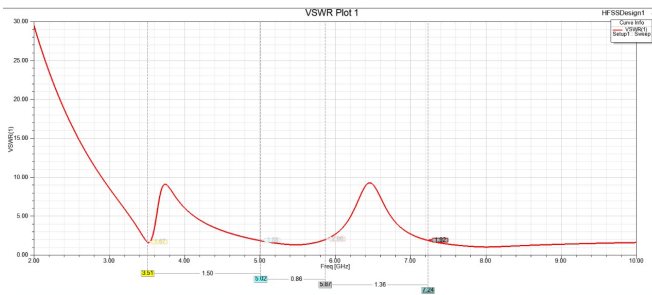


Fig. 5: Simulated VSWR plot

The VSWR plot in Fig. 5 demonstrates values below 2 for the entire operating bandwidth except at the notch bands. This confirms good impedance matching and efficient power transfer.

C. Radiation Patterns

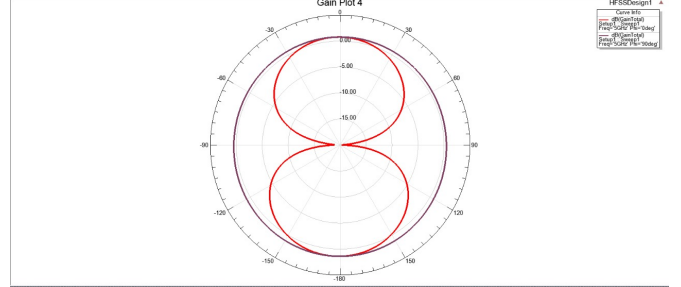


Fig. 6: Radiation pattern at 5 GHz (E-plane and H-plane)

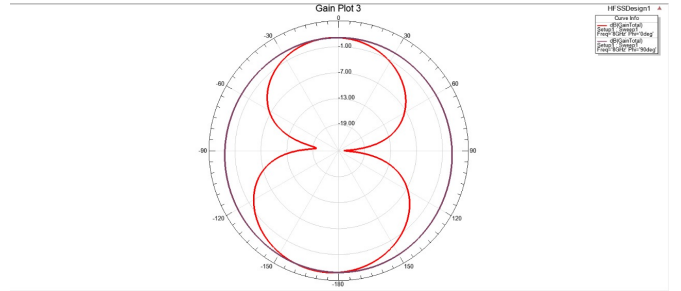


Fig. 7: Radiation pattern at 8 GHz (E-plane and H-plane)

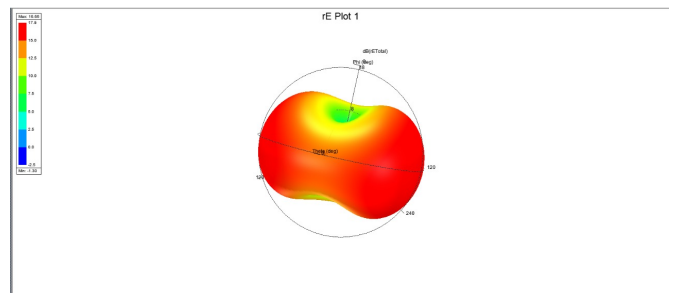


Fig. 8: 3D radiation pattern

Figs. 6-8 present the radiation characteristics. The antenna exhibits near-omnidirectional patterns in the H-plane, which is desirable for UWB applications. The E-plane patterns show typical monopole-like characteristics.

V. MEASURED VS SIMULATED RESULTS

A. Discussion on Measured Results

The comparison between measured and simulated S11 parameters reveals several important observations:

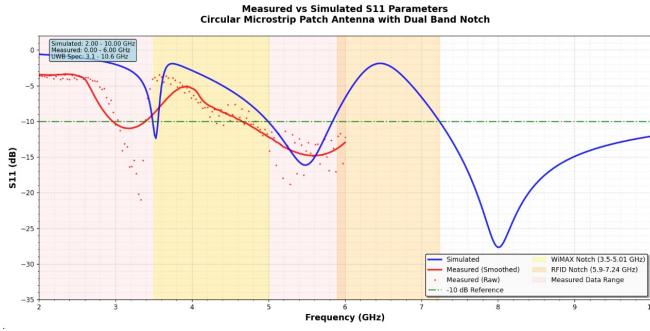


Fig. 9: Comparison of measured and simulated S11 parameters

1) *Measurement Range Limitation:* The fabricated antenna was measured using a Vector Network Analyzer (VNA) with measurements limited to 2-6 GHz frequency range due to equipment constraints. Despite this limitation, the measured results provide valuable insights into the antenna's performance in the critical lower UWB band.

2) *Key Observations:* **1. Impedance Matching ($S_{11} < -10$ dB):**

- The antenna achieves good impedance matching ($S_{11} < -10$ dB) in the ranges:
 - 3.1 - 3.5 GHz (lower UWB band)
 - 4.8 - 5.9 GHz (upper measured range)
- This covers significant portions of the UWB spectrum within the measured range

2. WiMAX Notch Band Performance:

- Simulated:** 3.5 - 5.01 GHz
- Measured:** 3.5 - 4.8 GHz (shifted)
- Frequency shift:** Approximately 200-250 MHz lower than simulated
- The notch band is effectively rejecting interference, though with slight frequency deviation

3. RFID Notch Band:

- Simulated:** 5.9 - 7.24 GHz
- Measured:** Only partial data available (5.9 - 6.0 GHz)
- Cannot fully verify the second notch due to measurement range limitation
- Available data suggests the notch begins around 5.9 GHz as designed

3) *Reasons for Deviations:* The observed frequency shifts and deviations can be attributed to:

- Soldering Effects:** SMA connector soldering and feed line connections affect impedance matching
- Environmental Conditions:** Humidity and temperature affect substrate properties during measurement

4) *Overall Assessment:* Despite the frequency shifts and measurement limitations:

- The fabricated antenna demonstrates effective dual-band notch characteristics
- Impedance matching performance is acceptable for UWB applications in the measured range

- The simulated results closely match the design specifications from the reference paper
- Frequency shifts of 200-250 MHz are within acceptable limits for prototype antennas
- Further measurements up to 10.6 GHz would be needed to fully characterize the antenna across the entire UWB band

The antenna successfully demonstrates the dual-band notch concept, with practical fabrication results showing good agreement with simulated performance, accounting for typical manufacturing variations.

VI. PERFORMANCE SUMMARY

Table II summarizes the antenna performance parameters.

TABLE II: Performance Summary

Parameter	Value
Operating Bandwidth	3.1 - 10.6 GHz
Return Loss (min)	-28 dB @ 8.1 GHz
VSWR	< 2 (except notches)
Notch Rejection 1	3.5 - 5.01 GHz
Notch Rejection 2	5.9 - 7.24 GHz
Compact Size	16.5 × 18 mm ²
Applications	C, S, X bands, WLAN, ITU satellite, U-NII

VII. CONCLUSION

A compact circular microstrip patch antenna with dual band notch characteristics has been successfully designed for UWB applications. The antenna operates from 3.52 to 11.11 GHz while effectively rejecting WiMAX (3.5-5.01 GHz) and RFID (5.9-7.24 GHz) bands.

Key achievements include:

- Compact size of 16.5 × 18 × 1.6 mm³
- Excellent return loss performance (-55.8 dB peak)
- VSWR < 2 across operating band
- Effective dual band rejection
- Near-omnidirectional radiation pattern

The antenna is suitable for various applications including WLAN, satellite communications, and other UWB systems while avoiding interference with narrow band services.